

the department of chemical engineering

vivek sharma

interim department head and professor



10.02.26

DEPARTMENT UPDATES



Thank you, **Vikas Berry**, for a decade of leadership.

(New building, faculty recruitment/retention/ concentrations, Saxena chair, growth in expenditures)



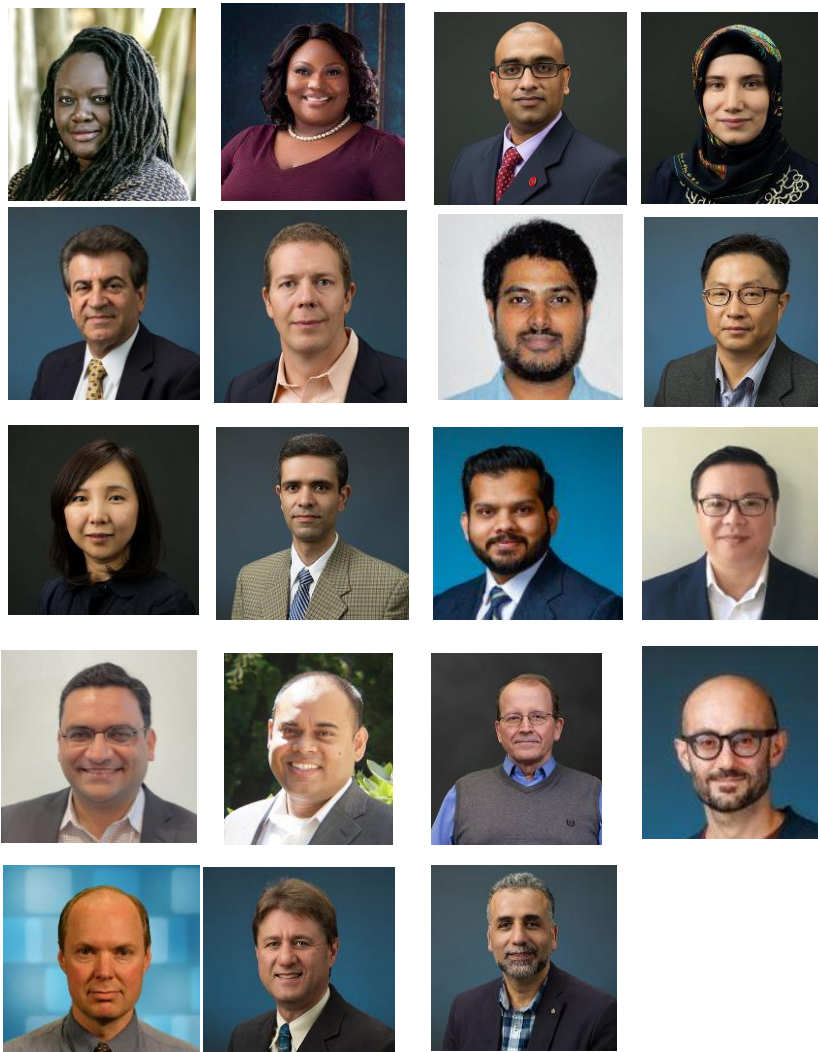
Professor, UIC | CEO, Argo Graphene Solutions | Founder, Grapherry | Building the Graphene Refinery

Welcome back, **Belinda Akpa**, as a professor with tenure

“The MADBio lab uses mechanistic mathematical modeling as a formal, testable sensemaking approach to navigate the path from extant biological knowledge to actionable insights.”



Growth in the number of faculty and students
Growth in research areas, funding, and expenditures



Faculty Members : 19

Tenure-Track : 12

Full : 8

Associate : 3

Assistant : 1

Clinical-Track: 4

Full : 1

Associate : 2

Assistant : 1

Research-Track : 3

Affiliated Faculty : 18



Sarai Chavez – Office Support Assistant



Meghan King – Assistant Director of Academic Communications



Eduardo Martinez – Grad./Undergraduate Program Coordinator



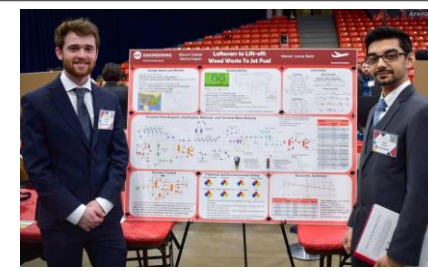
Jan Raphael Sagun – Laboratory and Computers Manager

Cassandra Wilson – Business Manager

Congratulations Jan on the
Staff Merit award



BS Degree Program	: 218 (188)
Graduate Program	: 54 (54)
MS Degree Program	: 10 (9)
PhD Degree Program	: 44 (45)
BS-MS Joint Program	: 4 (5)



Broadly our story so far

STORY

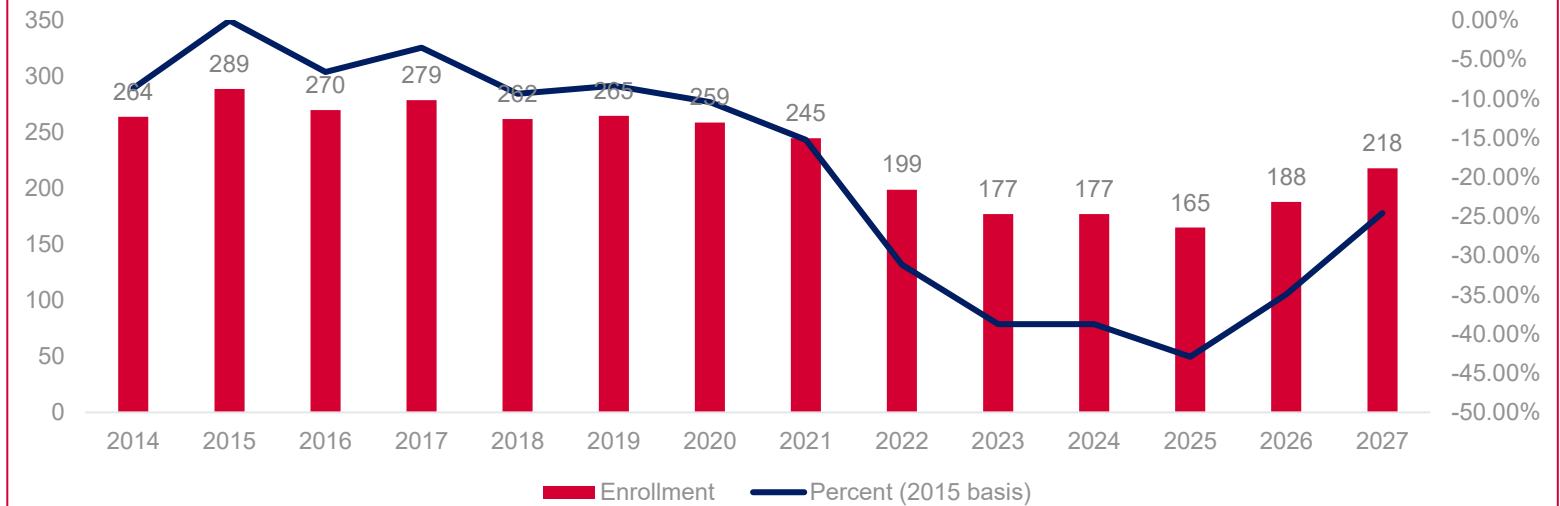


2019

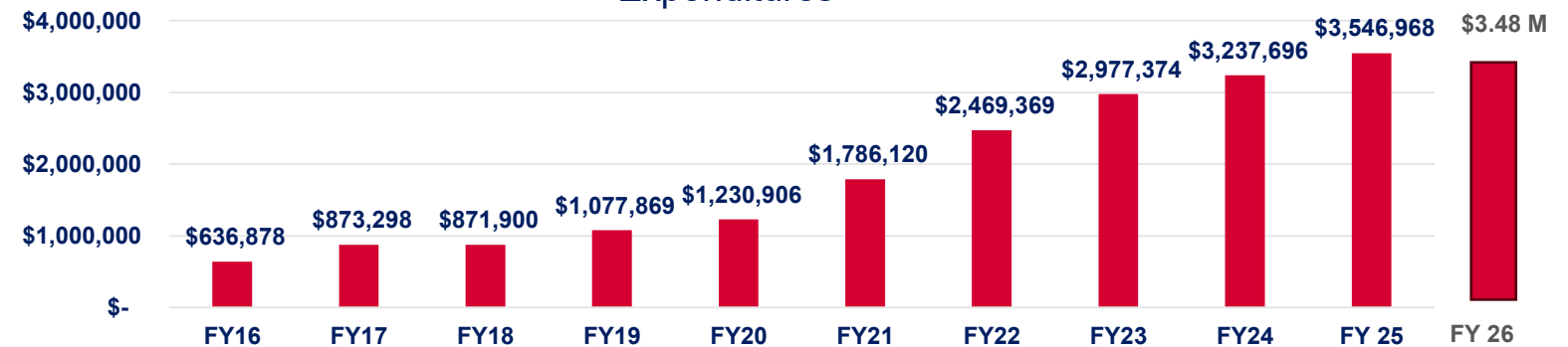


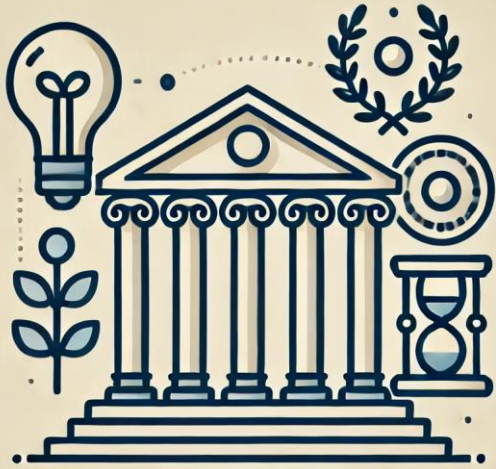
- State-of-the-art research laboratories
- Vibration-damped nanotech labs
- Energy labs: artificial leaf, fuel cells, solar cell batteries
- Special facilities: bioprocessing lab, dark room, ultrapure water, X-ray spectrometer room
- Enables hands-on student training across new concentration tracks

Enrollment



Expenditures





Department Rankings

- US-NEWS
 - Chemical Engineering (overall) = 62
 - Chemical Engineering (public) = 38
- QS Ranking
 - Chemical Engineering (overall) = 33
 - Chemical Engineering (public) = 20



STUDENT PLACEMENTS

Honeywell
UOP

DU PONT

TESLA

DOW

L'ORÉAL

bp

Exxon

BASF
The Chemical Company

Abbott
abbvie

JACOBS
NALCO
CABOT

Ingredion

GELITA
Improving Quality of Life

LA-CO

ECOLAB

AthenaVisual, Inc.

Ambitech

GATX

CBI

NAVISTAR

OROCHEM
Molecular Purification by Design

BAKER RISK

SIEMENS

ENEngineering

EXIDE
TECHNOLOGIES

SCP
South Chicago
Packing

Chobani

CARDINAL
INTELLECTUAL PROPERTY

Air Liquide

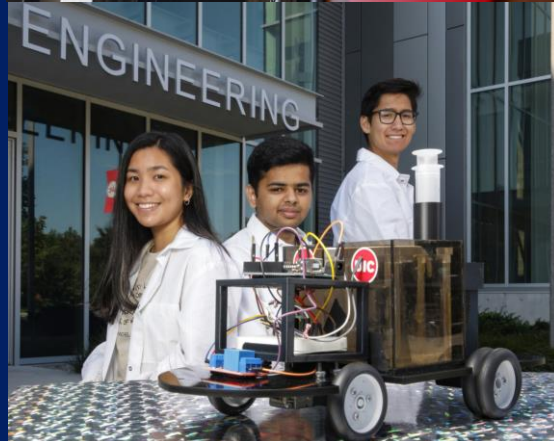
Cornerstone Controls

middough

CAMPOS
EPC

tpi
TheraProteins

Sargent & Lundy LLC



WORLD-CLASS
FACULTY MAKES A
WORLD-CLASS
DEPARTMENT

Entrepreneurship

Chemical Engineering Faculty and Students are collaborating with the following resources to translate ideas into products



UIC - Technology Innovation
Lab

**CHANCELLOR'S
TRANSLATIONAL
RESEARCH INITIATIVE**



13 faculty/student startups
3 CTRI grants received
2 POC grants received
1 DPI grant received
3 NSF-ICORPs
2 operations in mHUB

UIC Catalyst Center (UIC-CAT)

INITIATIVES



- Mission
 - Lead in industrial R&D
 - Translate lab innovations to market in collaboration with industry
 - Train industry-ready workforce
- Capabilities / Thrusts
 - Photo / Electro / Thermo / Bio-catalysis
 - Reactor design, scale-up & pilot testing
 - High-throughput characterization & AI modeling
 - Shared access: UIC + Argonne facilities
- Industry value & asks
 - Value: de-risk tech · talent pipeline · early licensing access
 - Asks: 1) Internships · 2) Seed-sponsored projects · 3) Join Advisory Board / co-develop roadmaps

UIC-CAT Working Groups

- Working Group 1 – Sustainable and Novel Chemicals, Materials, and Processes (SusChemMaP)
 - Ksenija Glusac (Photochemistry), Brian Chaplin (Electrochemistry), Neal Mankad (Thermochemistry)
- Working Group 2 – Design, Prototyping, Advanced Manufacturing, and Process Scale up (DePAMS)
 - Yayue Pan (Prototyping), Yuepeng Zhang (Advanced Manufacturing), and Constantine Megaridis (Process Engg & Scale up)
- Working Group 3 – Advanced Artificial Intelligence and Multiscale Modeling for Digital Twins (AMD)
 - Sara Kadkhodai (Molecular Simulations), Santanu Chaudhuri (Multiscale Modeling), Michael Papka (Artificial Intelligence)
- Working Group 4 – High Throughput Experiments and Multimodal Characterization (HiTEC)
 - Ahmed Farghaly (High Throughput Experiments), Robert Klie (Multimodal Characterization), Reza Shahbazian-Yassar (Microscopy)



UIC-CAT Industry Partners

**Kushal Sinha**

Founder & CEO, Kortana AI

Cognitive twins for scientific R&D;
Former Principal Scientist at AbbVie; Ph.D., UW–Madison;
AIChE 35 Under 35.

**Antarip Halder**

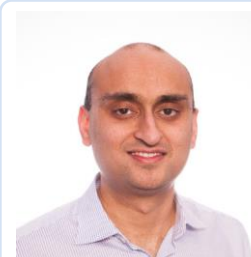
Senior Scientist, Aganitha AI

Leads AI-augmented molecular modeling platforms integrating
QM, MD, and ML for drug discovery and materials design.

**Eric Murphy**

Senior Scientist II, AbbVie

Digital twin and computational multiphase modeling initiatives for
high-throughput experimentation and process platforms.

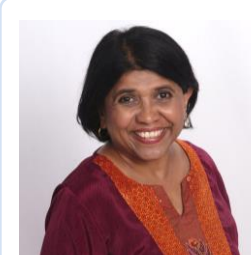
**Ankur Kapil**Technology Leader, Pharma Process
Development

Expertise in ML-driven process design, crystallization, spray drying,
and process development; Ph.D., University of Manchester.

**Gian Prakash**

Global Head, Data & Analytics, AbbVie

Leads AI-driven digital infrastructure for clinical-trial operations
using knowledge graphs and advanced analytics.

**Urmila Diwekar**

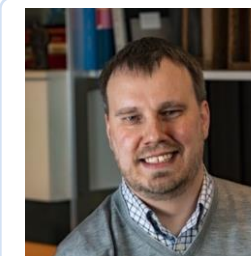
President, Vishwamitra Research Institute

Globally recognized expert in optimization under uncertainty;
AIChE Fellow; recipient of multiple national awards.

**Udit Gupta**

Technical Account Manager, Siemens

Specializes in process modeling and digital twins;
Ph.D., University of Minnesota.

**Andrey Ivankin**

Co-founder & CTO, Mattiq

Accelerates materials discovery using data-driven and high-
throughput approaches; Ph.D., Illinois Institute of Technology.





Ilke Arslan

Scientist, Argonne National Laboratory

Advanced characterization and high-throughput approaches for materials and catalysis.



Nan Chen

Senior Researcher, Chevron

20 years of experience in catalysis, refinery optimization, and high-throughput experimentation systems.



Swati Neogi

Professor, IIT Kharagpur

Works on machine-learning approaches for advanced materials and composites.



John Finley

CEO, Nanofixation

Entrepreneur with finance, commercialization, and technology-venture experience; translates research into market impact.



Saurabh Chaugule

Technology Leader, AWS

AI-driven transformation across industrial sectors; expertise spanning energy, data platforms, and applied analytics.



Vimal Srivastava

Professor, IIT Roorkee

Faculty member focused on AI/ML applications in chemical industries and chemical engineering systems.

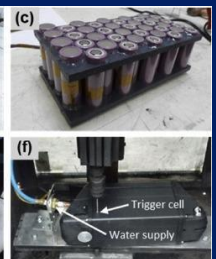
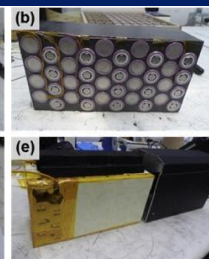
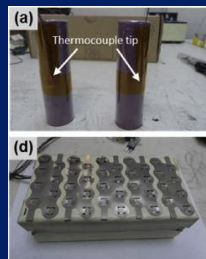
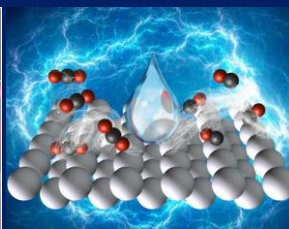
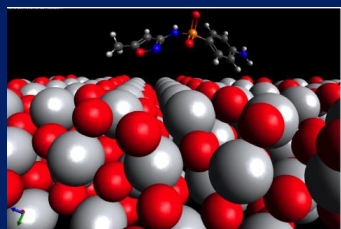
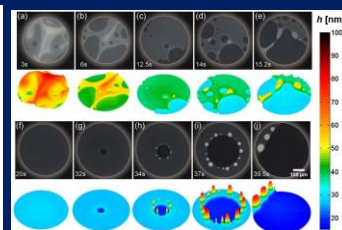
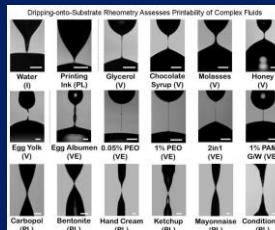
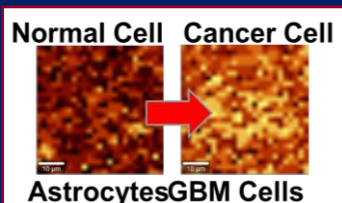
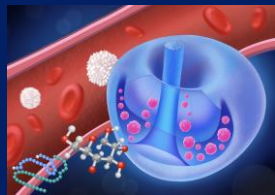


Opportunities for Industry Members: UIC-CAT

- **Project Feedback** – Industry members can provide direct input to shape the direction of UIC-CAT research projects to align with real-world industrial needs.
- **Access to Fundamental Research Capabilities** – Members gain access to UIC’s multidisciplinary expertise in catalysis, enabling collaborative solutions to complex technical challenges.
- **Access to Recruiting Pool** – Industry partners can engage with and recruit top-tier undergraduate, graduate, and postdoctoral talent from UIC.
- **Access to UIC Instruments and Equipment**– Members benefit from discounted access to UIC’s advanced research instruments and facilities for testing and analysis.
- **Collaboration with ANL Scientist through Crabtree Institute** – Through UIC-CAT’s partnership with Argonne National Lab, members can access national lab capabilities via the Crabtree Institute.
- **Access to Patents for Licensing** – Members receive early visibility and licensing opportunities for catalysis-related intellectual property developed at UIC.
- **Possibility of Teaching Students** – Industry professionals can contribute to education by guest lecturing, mentoring, or co-developing course content with UIC faculty.
- **Inclusion of Industry-Driven Workshops** – Members help shape and participate in targeted workshops and symposia that address critical challenges and innovations in catalysis.

Chemical Engineering

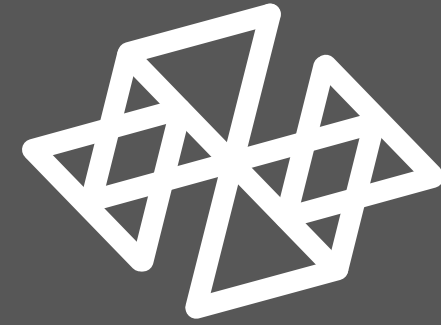
OTHER OPPORTUNITIES



Strategic Research Areas (industry relevance)

- Energy — batteries, fuel cells, solar, artificial leaf (scale-up & materials)
- Nanotechnology — advanced materials, coatings, sensors
- Biomedical Devices — diagnostics, therapeutics, point-of-care systems
- Water Science — contaminant separation & treatment processes
- AI / Data Science — process optimization, predictive maintenance
- Advanced Manufacturing — pilot-scale prototyping, process intensification
- Sustainable Foods & Biomass — plant-based ingredients, conversion tech
- Complex Fluids & Molecular Engineering — formulations, separations
- Pharmaceuticals Engineering

Modeling for Accelerated Discovery in Biology (MADBio) Lab



The MADBio lab uses mechanistic mathematical modeling as a formal, testable sensemaking approach to navigate the path from extant biological knowledge to actionable insights.

1

Modeling human immune responses for accelerated design of immunogens that provide broad and durable pathogen protection

2

Model-driven discovery and interruption of the processes that drive chronic immune responses and the failure of biomedical implants

3

Strategic dissection of systems responses to nutrient stress in plants to inform the next generation of interventions for agricultural resilience

4

Quantifying and establishing to decision-making consequences of prediction uncertainty in model-driven design of therapeutics

Research Leadership (examples): (1) Computational core of the Humanity Unlocking Biomaterials (HUB) NIBIB U24 Biomaterials Coordinating Center (2) National Cancer Institute S³ (synthetic, systems, and spatial) Innovation Lab (3) Department of Energy Office of Science Pandemic Response Roundtable

MoDSE Lab: Research Overview

Molecular Design for Sustainable Engineering | PI: Ezinne C. Achinivu-Ibagere, Ph.D., Assistant Professor of Chemical Engineering

We develop molecular design principles for transforming waste (biomass, plastics and agricultural residues) into valuable products, through three integrated thrusts plus a growing AI/ML-guided design effort.



01

Green Chemistry & Catalysis

- Protic ionic liquids for biomass deconstruction and upgrading
- Metal–ionic liquid framework (MIF) catalysts for selective lignin depolymerization
- Lignin to fuel-range cycloalkanes



02

Advanced Biomaterials

- Structure–property–processing design of lignocellulose-derived materials
- Cellulose–lignin bioplastics as LDPE alternatives
- Biodegradable soy-based agricultural films



03

Biotechnology & Bioconversion

- Robust microbial communities that turn waste into bioproducts via anaerobic digestion
- Enzyme–solvent compatibility
- Bioactives from food waste (sinapic acid)



04

AI-Guided Solvent Design

- Molecular modeling and machine learning coupled with experiments
- Task-specific solvents for biomass and plastic waste
- Critical mineral separations



Engineering Education Research



BETUL BILGIN

Bridging
Academia & Industry

Engineering Identity
Development



Industry-relevant,
contemporary problem
design and implementation

Diversity
Equity and
Inclusion

DEI faculty and student
committees, Climate Surveys,
Mentorship Programs

Sustainability,
Gen-AI and Data Science
Integration

Data Science module design
and implementation, CHE 101
Design, Sustainability activities

K-12 Outreach and
Interest in Chemical
Engineering

COE Summer Camp,
Engineering Identity of K-12
students, Water marbling

Senior Design I/II

Team-based Learning

Intro to Thermodynamics

Active Learning

Computational Methods

Problem-based Learning

Transport Phenomena III

Computer Simulation

AICHe 35 Under 35

ASEE 20 under 40

UIC TRP Award

UIC COE Harold Simon

ASEE CHED Ray Fahien

Environmental Electrochemistry Lab: Research Overview



CHEMICAL
ENGINEERING

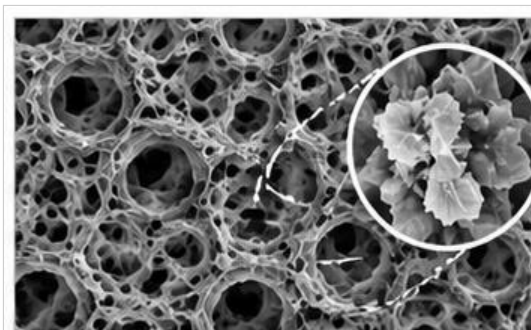
Brian P. Chaplin, Ph.D. Professor & Director of Graduate Studies, Chemical Engineering, University of Illinois Chicago

We develop electrochemical and membrane-based technologies for sustainable water and resource recovery, with an emphasis on fundamental understanding of interfacial reactions, advanced materials, and scalable treatment



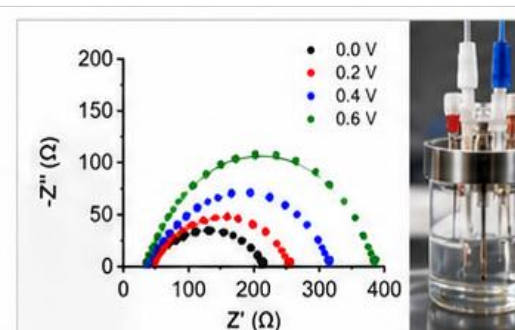
Electrochemical Water Treatment

- Electrochemical oxidation and reduction for contaminant destruction
- Reactive electrochemical membranes & flow-through reactors
- Removal and destruction of PFAS, nutrients, & emerging contaminants



Electrode & Membrane Materials

- Design of high-performance electrocatalytic materials
- Structure–function relationships for interfacial reactivity and stability
- Scale-up and manufacturing of reactive membrane assemblies



Interfacial Electrochemistry & Mechanisms

- Reactive transport modeling
- In situ characterization (e.g., EIS, spectroscopy)
- Molecular modeling to interpret experimental data & determination of reaction mechanisms



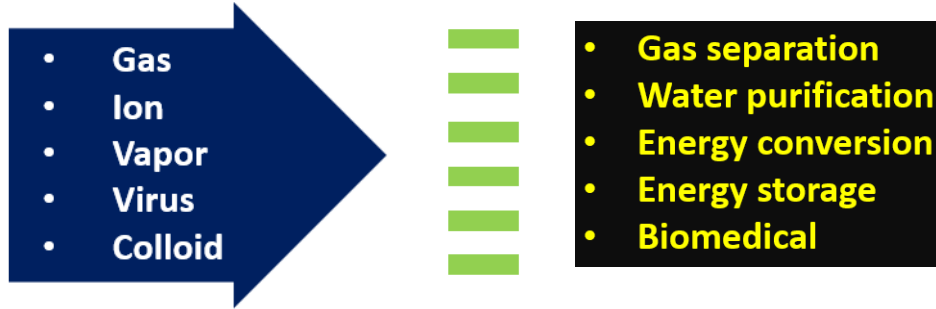
Water Circularity & Implementation

- Electrochemical recovery and production of value-added products
- Integration with renewable energy and low-carbon systems
- Technoeconomic and life cycle assessment for real-world implementation

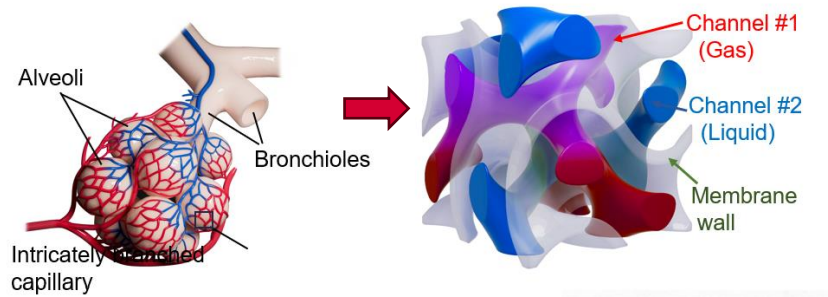


Group Research Theme (S. Kim)

□ Nanoporous Materials for Molecular Transport



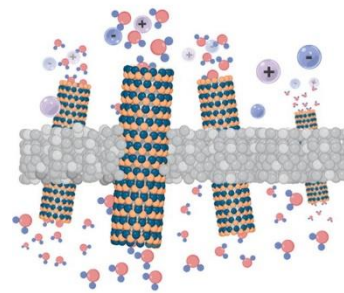
□ 3D-printed Biomimetic Membranes



- High-performance membrane contactor for CO₂ capture
- Artificial kidney
- Compact blood-to-plasma separation filter

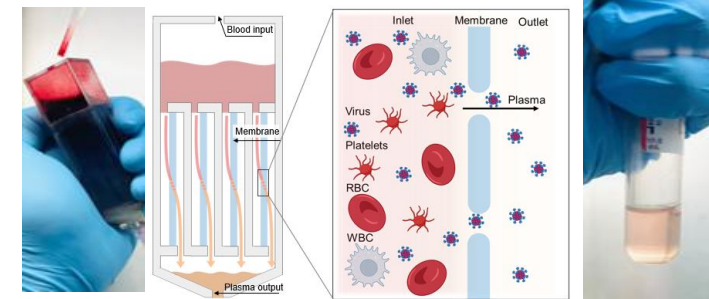
□ Water Purification/Energy Conversion

Vertically aligned BNNT nanofluidic platform



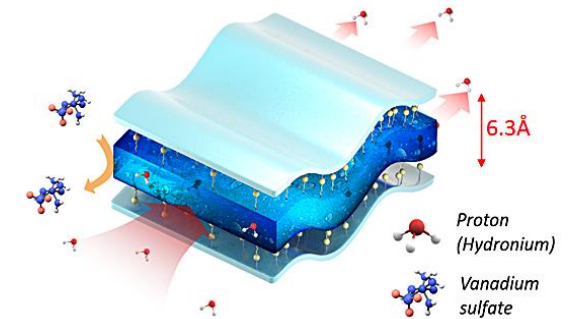
- Water purification
- Green energy production

□ Bioparticle Separation

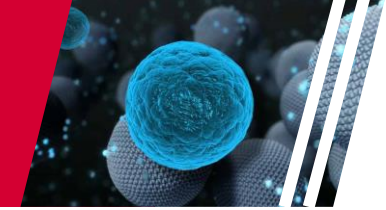


- Virus separation and purification
- Blood sample preparation

□ Electrochemical Membranes



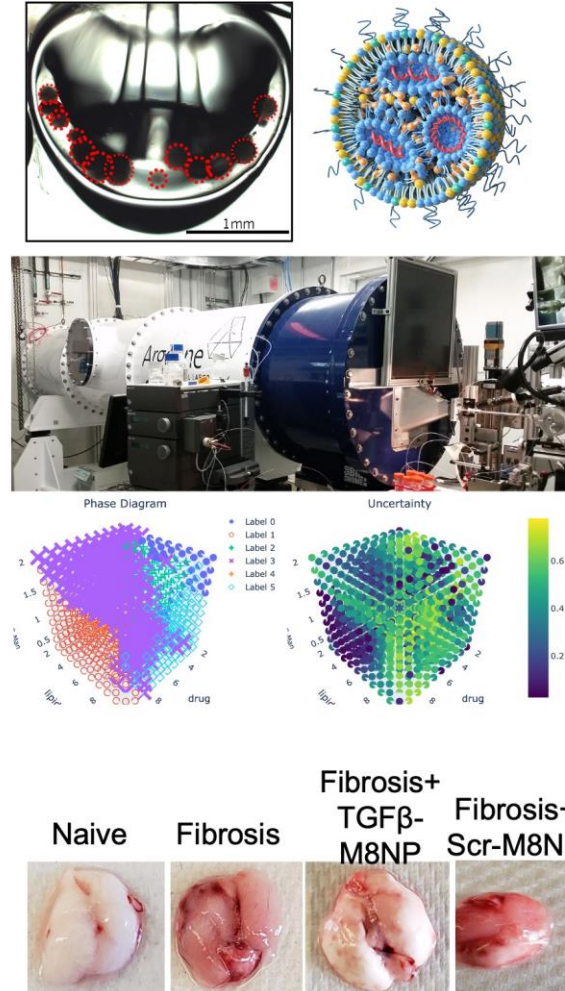
- Redox flow batteries
- All-solid-state electrolyte batteries
- Bipolar membrane



- AI-guided autonomous platforms integrating high-throughput flash mixing and advanced characterization for lipid nanoparticle (LNP) optimization and structure–function relationship discovery
- AI-guided tangential flow filtration systems for continuous separation and purification
- Advanced delivery systems for small-molecule drugs, antibodies, peptides, nucleic acids, and therapeutic cells
- Synchrotron X-ray and neutron scattering to investigate interfacial biophysics, including biomolecular packing, interactions, organization, and reactions
- Microfluidic systems and complex fluids

Sanofi iDEA-TECH award, 2026

Sino-American Pharmaceutical Professionals Association
Leadership Award, 2023, 2026



Molecular Simulations Laboratory (Mehraeen's Research Group)



- Electrochemical Reaction Mechanisms

Computational modeling of electrochemical oxidation mechanisms for water contaminants, revealing how pollutants break down at the molecular level.

- Electrocatalyst Design

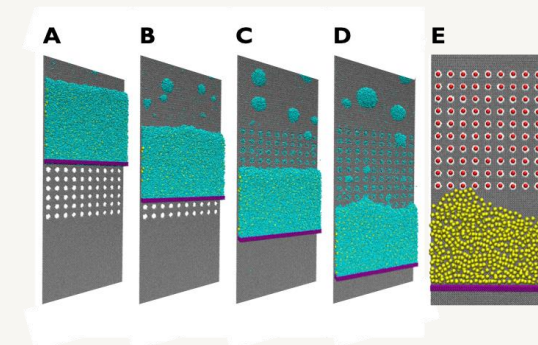
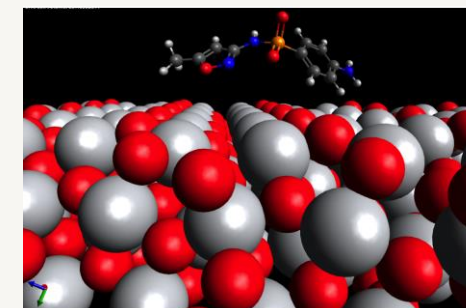
Translating mechanistic insight into rational design of next-generation electrocatalysts for water treatment.

- AI-Automated X-ray Scattering

Using AI to automate X-ray scattering experiments at the Advanced Photon Source (Argonne National Laboratory) to accelerate catalyst discovery.

- Digital Twins for Lipid Nanoparticles

Building digital twin models to predict lipid nanoparticle properties, guiding the design of next-generation drug delivery systems.



Anh Ngo Research Group | Research Overview

Computational and experimental design of materials for next-generation energy · UIC Chemical Engineering



1

Next-Generation Batteries

Designing anodes, cathodes, electrolytes, and the interfaces between them for safer, higher-energy storage.

Li-metal · Na-ion · solid-state · interfaces · coatings · ion transport



2

Multiscale Materials Modeling

A quantum-to-continuum pipeline that links atomic-scale physics to device behavior, validated against experiment.

DFT / TDDFT · classical & ab-initio MD · ML force fields · continuum



3

AI + Quantum Optimization

Physics-informed ML and quantum-annealing solvers for large combinatorial energy-system problems.

QUBO · quantum annealing · physics-informed ML · microgrids



4

Energy & Functional Materials

Catalysts, photovoltaics, and strongly-correlated quantum / critical materials for sustainable technology.

catalysis · graphene-oxide PV · quantum / critical materials



\$1.86M

in new research awards (2025–26)



4

sponsored projects launched



DOE · Argonne · Industry

funding partners

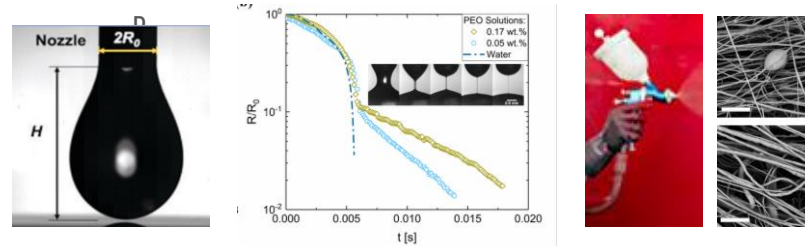
Quantum → atomistic → mesoscale → device — integrated with machine learning and experiment

Sharma's Soft Matter ODES-Lab: Plant-based Foods & Sustainable Formulations



Polymer Physics & Rheology

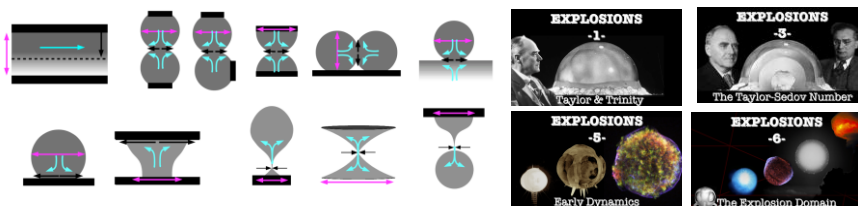
Stretched polymer physics. Extensional rheology



Spray Drift (BASF) & Particles in Viscoelastic (VE) Fluids (PPG)
Non-Newtonian fluid mechanics, fiber science (BASF, 3M)

Scaling & Dimensional Mechanics

Spreading, Pinching & Coalescence; Explosions



Mechanics: key length scales, timescales, speeds:
<https://www.youtube.com/@naturesnumbers/about>

Molecular Engineering of Formulations

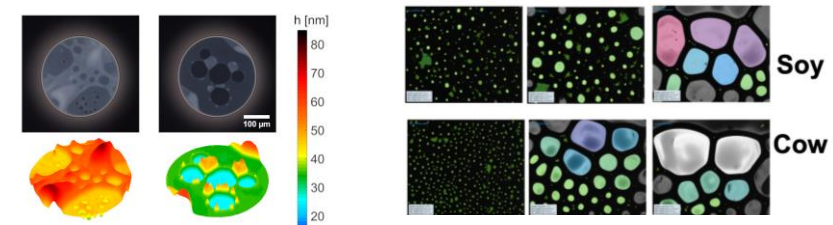
Sustainable Formulations; Plant-based foods (Milk, Mayo)



Plant-based proteins for Milk & Mayo (Motif); Stringy yield stress fluids (Kraft-Heinz); Foamability, Emulsion stability

Fizzics (Drops, Bubbles, Foams, Emulsions)

Polymer-protein complexes, Bio-interfaces, Surface forces



ODES-lab students, alumni, publications:
<http://viveksharmalab.com>

We pursue curiosity-driven, bio-inspired, and industry-motivated research problems.

Collaborations with PPG, 3M, BASF, Motif Foodworks, Dow, Colgate, Kraft-Heinz, UIUC, Sorbonne, Purdue, CNRS, Northwestern, Case Western, U Houston

Alumni: 3M, P&G, AbbVie, Abbott, Honeywell, People Gas, BP, Mars, MedCision, Diageo, NALCo Ecolab

MIT, Stanford, Berkeley, UIUC, U. Chicago, UCLA, Wisconsin-Madison... & Med school

Material & Systems Engineering Lab (MaSEL)



105+

Peer-Reviewed
Publications

8

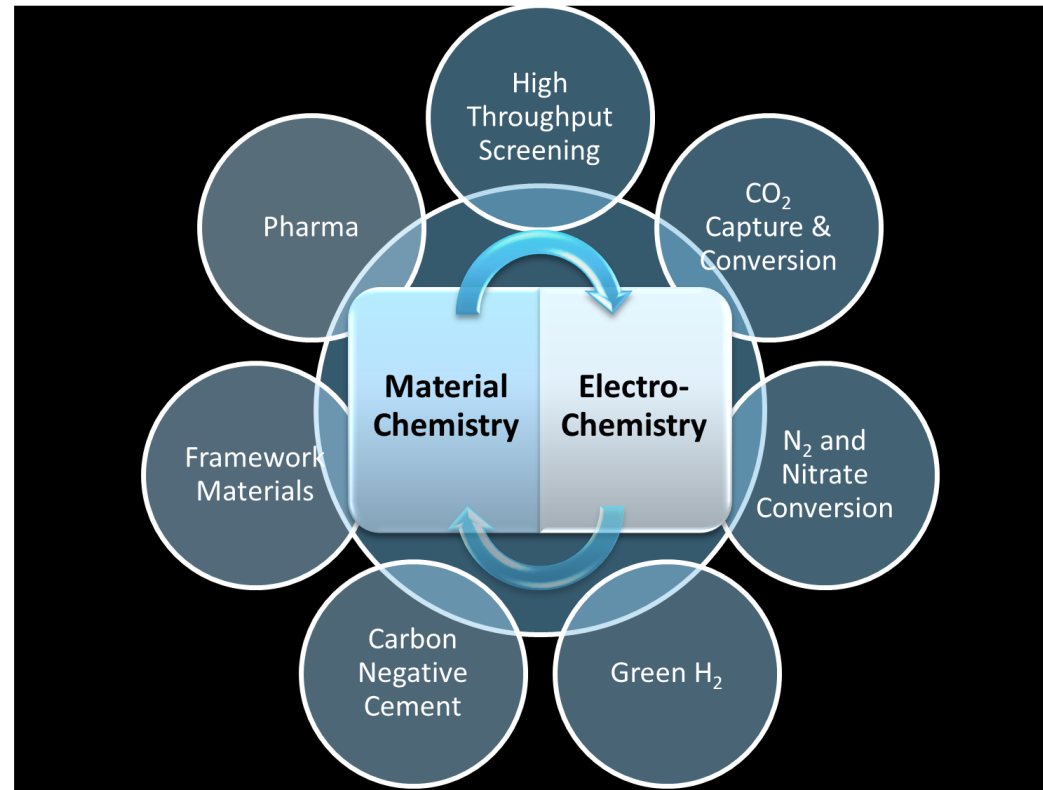
Patents

3

Licensed
Techs

4

Book Chapters



NSF Grants –
EFRI, ECO-
CBET, CBET,
CMMI

DOE Grants– BES and
ARPA-E

Industrial Grants–
Braskem, Saudi
Aramco, eN-RAMPS,
General Ammonia,
AbbVie, BMS, Takeda,
Eli Lilly, Biogen.

Chemical Engineering at a Glance

19

Faculty

3 NSF Career Awardees

282

Students

1

Undergraduate
Degree Programs

1

Master's Programs

11

Research Labs

1

Interdisciplinary
Center (coming up)

\$3.5M

FY26

Expenditures

approximately

5

staff

QS
Ranking

#20

US Public

TOP 33

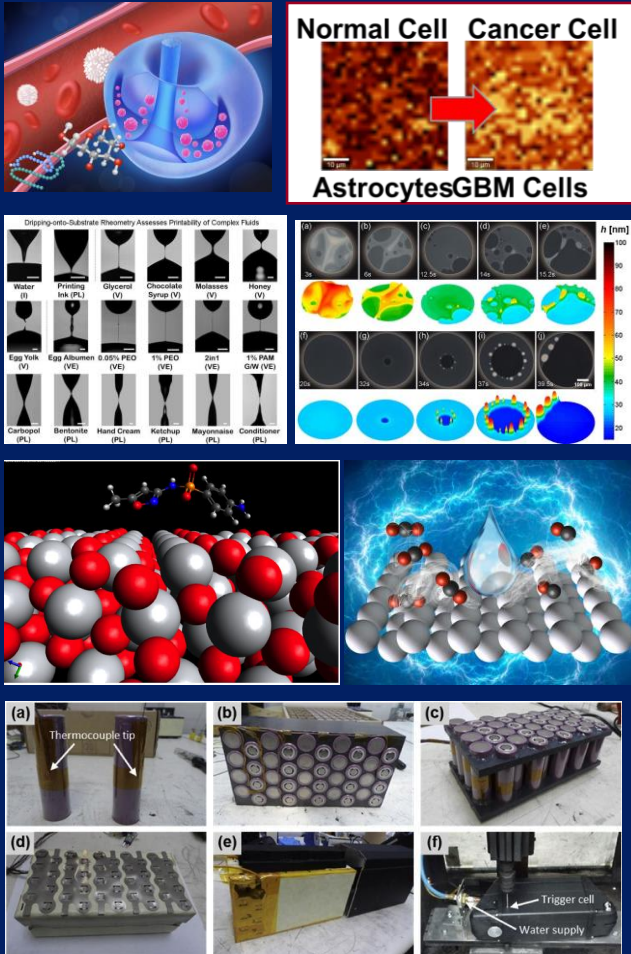
US

THANKS



Chemical Engineering

ENGAGEMENT OPPORTUNITIES

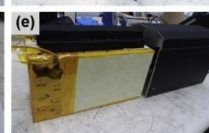
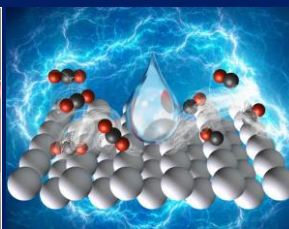
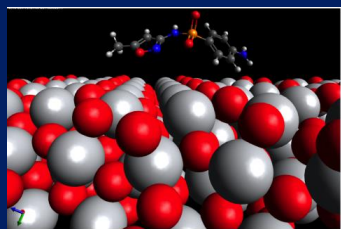
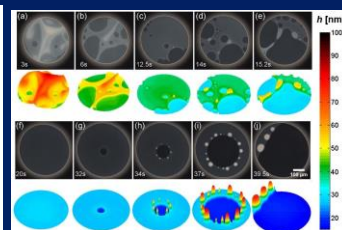
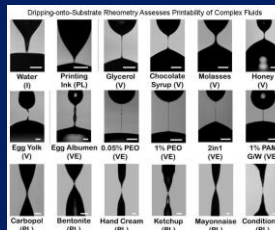
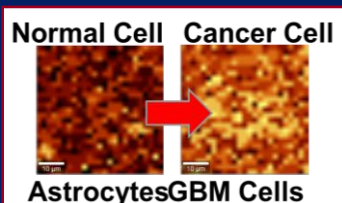
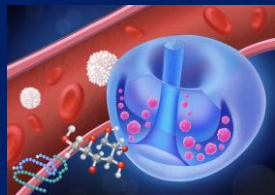


Legacy Tier

- \$500K — Named Professorship
 - Establish a named professorship to support salary and research startup needs.
 - Visible recognition: named title, web & event recognition, annual partner briefing.
 - Impact: attracts/retains senior faculty, seeds focused research & student mentoring.
- \$2M — Chaired Professorship
 - Endows a chaired professorship with full recurring support for salary, research, and graduate fellowships.
 - Premier recognition: named chair, prominent campus visibility, invitation to strategic advisory events.
 - Impact: guarantees long-term leadership in a priority area, drives sustained industry-relevant research and workforce development.

Chemical Engineering

ENGAGEMENT OPPORTUNITIES



Collaborate Tier — \$25–75K

Typical commitments:

- Sponsored seed research project (\$25–75K, 6–18 months)
- Co-branded event (innovation challenge, training workshop)
- Multi-internship cohort or co-op support (stipends for 3–5 interns)
- Large equipment purchase (unit operations equipment)

Benefits to partner:

- Project milestones and quarterly technical/reporting meetings
- Priority access to student project teams and candidate interviews
- Co-branding on materials + dedicated project page and case study

Partner Tier — \$100K+

Typical commitments:

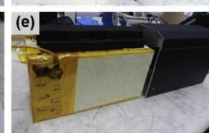
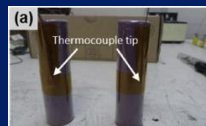
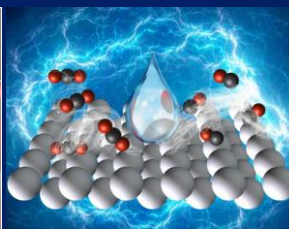
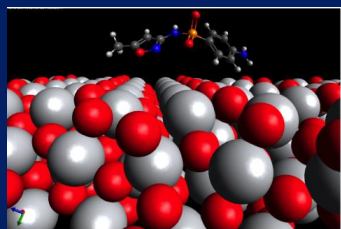
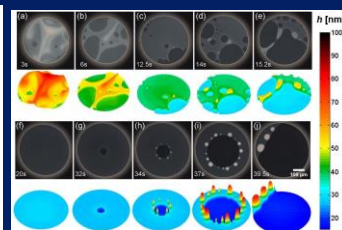
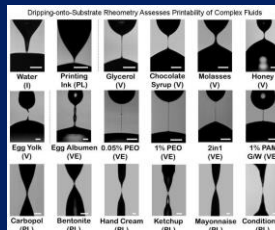
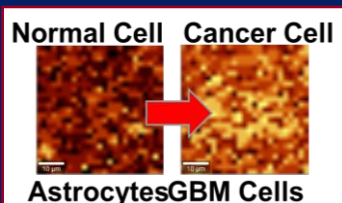
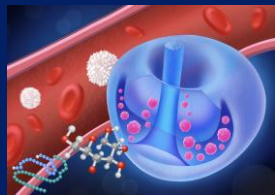
- Multi-year sponsored research program (>\$100K/year)
- Dedicated graduate fellowship or multi-year internship program
- Equipment co-funding for pilot or shared facility
- Facility renovation / dedicated lab sponsorship

Benefits to partner:

- Named project or program page, annual impact report with KPIs
- Dedicated faculty liaison
- Customized recruiting events + priority access for hiring

Chemical Engineering

ENGAGEMENT OPPORTUNITIES



In-Kind Tier (Non-cash / Entry)

Typical commitments:

- Student Mentors
- Industry site visits, plant tours, access to test data
- Classroom Instruction or Guest Lecture/Workshop
- Equipment loan or donation (benchtop instruments)
- Small in-kind scholarship (e.g., travel grants for students to conferences)

Benefits to partner:

- Acknowledgement on website & event materials
- Invitation to department events; priority on student visits

Engage Tier — \$5–10K

Typical commitments:

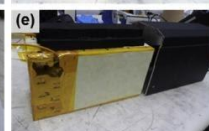
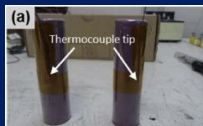
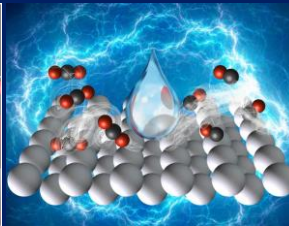
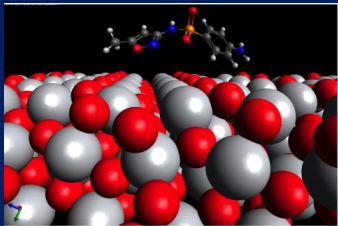
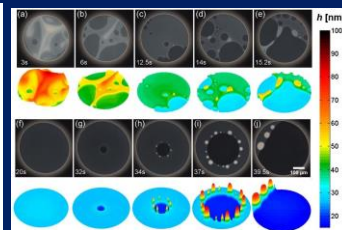
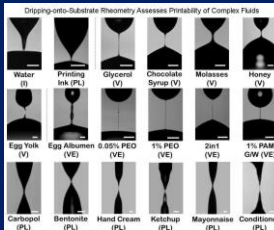
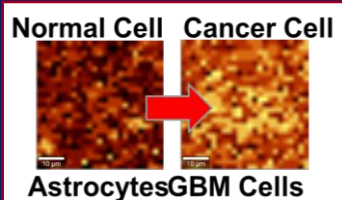
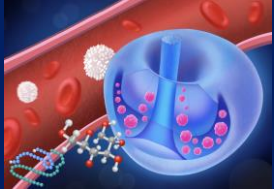
- Sponsor one senior capstone project (mentor + \$2k-\$5K materials)
- Chem-E-Car or Chem-E-Cube support (\$8k)
- Equipment donation (< \$10K)
- Hire 1–2 paid summer interns and/or fund an internship stipend
- Student travel grants (~\$3k each)
- Computers (\$2.5k each)

Benefits to partner:

- Logo on capstone materials + “Engage Tier” listing on site
- Candidate shortlist for internships
- Panel or guest lecture opportunity

Chemical Engineering

OTHER OPPORTUNITIES



Unit Operations Experiments

- Reaction Engineering Experiment: \$40k
- Biochemical Reactor: \$40k
- UV-Vis Spectrometer: \$15k
- Viscosity upgrade: \$5k

Computers

- Equipment: \$2.5k each

Student Travel Grants

- \$3k each

Chem-E-Car

- Equipment+Supplies: \$8k
- Chem-E-Car Fellowships: 3 X \$3k = \$6k

Chem-E-Cube

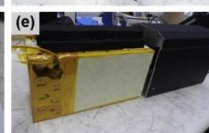
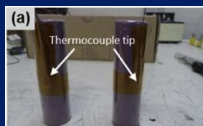
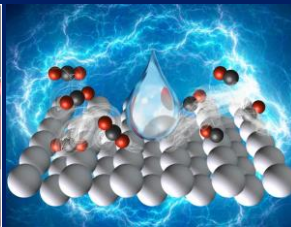
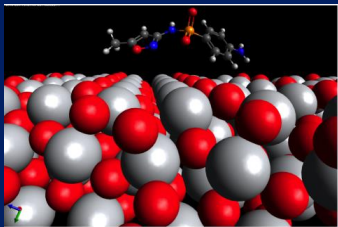
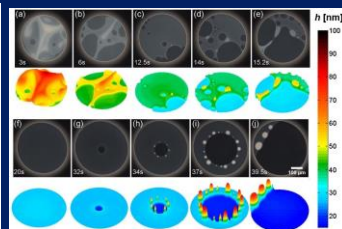
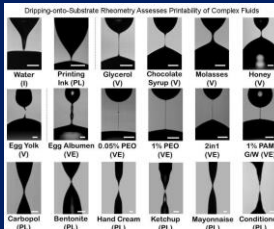
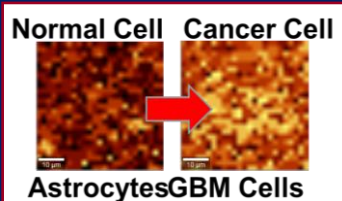
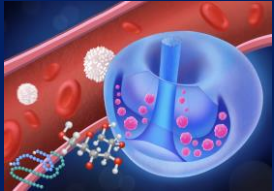
- Equipment+Supplies: \$8k
- Chem-E-Cube Fellowships: 3 X \$3k = \$6k

THANKS



Chemical Engineering

ENGAGEMENT OPPORTUNITIES



Industry Partnership & Talent Opportunities

- Internships & Co-ops: paid placements in pharma, energy, materials
- Research Collaborations: sponsored projects, SBIR/TEAMING, joint labs
- Capstone & Problems: real company problems for senior design teams
- Faculty Start-ups & Licensing: early access to spinouts
- Facility Access: pilot testing, characterization, prototyping in EIB

How You Can Engage — Concrete Asks

- Sponsor 1–2 paid interns next summer
- Submit 1 capstone problem
- Fund a \$50–150K seed project to de-risk promising tech
- Host a site visit or short industry challenge

Engineering at a Glance

230*

Faculty

50 NSF Career Awardees

6,398

Students

6 Departments

13

Undergraduate
Degree Programs

12

Master's Programs

150

Research Labs

4

Interdisciplinary
Centers

\$167M

FY24 Active
Sponsored Research

approximately

90

full-time staff

UIC - U.S. News
& World Report
Highlights

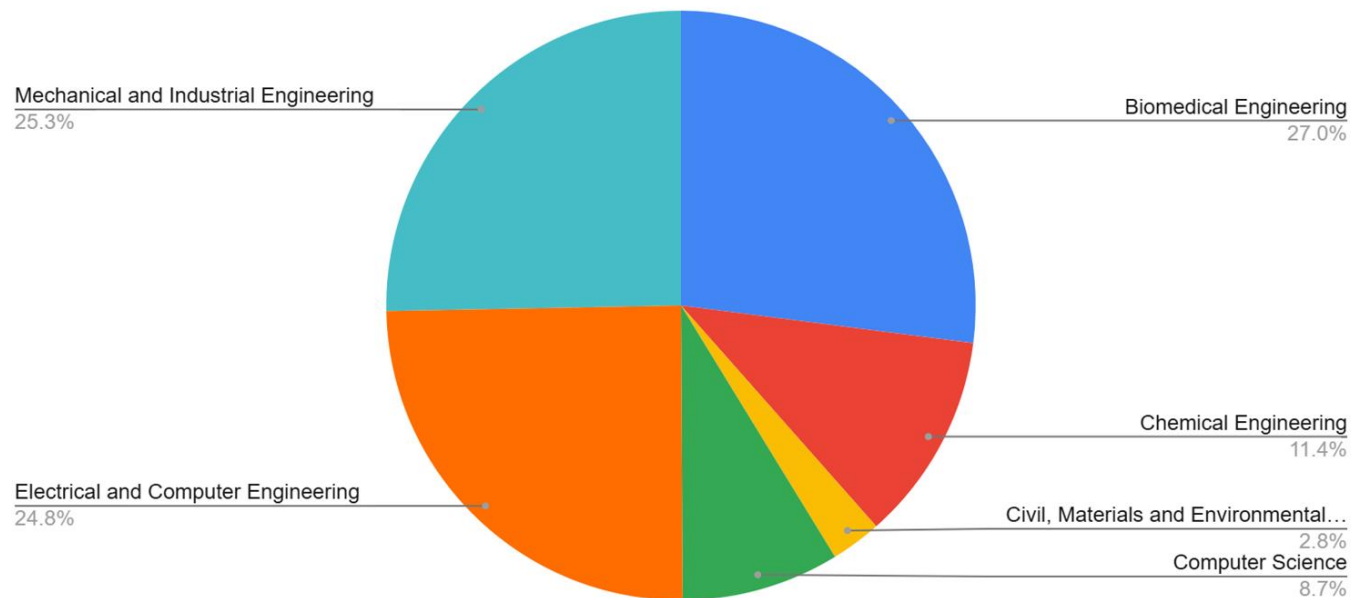
#10

Social Mobility

TOP 37

Ethnically
Diverse

Engineering Disclosures 2016-2025



Total for COE
 577 Disclosures
 76 Active Patents
 34 Active Licenses

FY	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025*
Disclosures	54	50	57	49	66	62	42	36	45	43
Patents Filed	44	32	38	37	23	37	38	36	38	20
Patents Issued	6	2	8	5	5	7	7	4	5	7
Licenses & Options	6	10	0	5	3	4	1	0	5	2

FY2025 ends 6/30/25